

a11oy.UDS

A UDS-native, governed agent runtime.
Inheriting UDS guardrails. Carrying a11oy's orchestration DNA.

Why this name, why this shape

The Linux ethos thread: the kernel surface stays small, sharp, and verifiable.
Capabilities mesh in at the policy and admission edges.

a11oy is a capability. UDS is the kernel surface.

What a11oy already carries

1 Orchestration plane signal mesh, planner, workcells	2 Approval gates no material action without human OK	3 Artifact registry models, prompts, embeddings, evals, agents
4 Proof ledger append-only, Ed25519 + ML-DSA-65	5 Λ-9 invariant runtime Doctrine V6 — 0.90 / 0.95 / 0.95	6 Recalibration memo weekly 'what changed' feed

What UDS already carries

01 Distribution Zarf packages, UDS bundles	02 Cluster uds-core, Pepr admission, NetworkPolicies	03 Identity Keycloak SSO, tenant realms
04 Edge Istio tenant gateway	05 Telemetry Loki, Prometheus, Grafana	06 Distribution at scale OCI registry + SBOM flow

The meshing thesis

a11oy primitive	UDS primitive it inherits from
Orchestration	Pepr operators, Istio tenant gateway
Approval gates	UDS policy engine, Pepr admission
Artifact registry	Zarf packages, OCI registry, SBOM flow
Proof ledger	Loki + signed attestation sidecar
Λ -9 invariant	Pepr admission module (#5027, merged)
Recalibration memo	NetworkPolicy-aware cluster inventory feed

Trusted AI/agent orchestration inside air-gapped UDS

- Strong image / chart provenance. ✓
- Strong identity (Keycloak). ✓
- Strong policy (Pepr). ✓

Governed agent runtime with structural approval gates,
immutable tool-call audit, and verifiable offline. ← gap

A UDS-native artifact spine for AI

Container images: signed, versioned, attested, lifecycled. ✓

Models, prompts, embeddings, agent definitions, evals:

- SBOM-style attestation per artifact

- Signed evals (dev pass means prod pass)

- Drift detection

- Promote / queue / discard, mirroring Zarf

A / B / C — one ladder, one direction

OPTION A 2-3 week proof point a11oy.UDS as a Zarf bundle payload. Drops into an existing UDS cluster. Inherits everything as-is.	OPTION B Falls out of A Primitives ported one-by-one to native UDS components. Done as a by-product of the bundle work.	OPTION C The real destination Full a11oy.UDS ecosystem port. First-class peer of uds-core. Default checkbox at install.
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The 2–3 week proof plan (Option A)

WEEK 1 Bundle, identity, observability • uds-cli bundle deploy on the reference cluster • Keycloak SSO round-trip end-to-end • Istio tenant gateway routes verified • Loki + Prometheus exporters confirmed	WEEK 2 Approval gates, Λ-9 admission, audit chain • Bad invocation → MATURITY_GATE_BLOCKED • Good invocation → approval queue prompt • Cable pulled — three offline invocations • verify --offline → OK chain=clean	WEEK 3 Artifact spine • AIArtifact CRD applied • All five kinds round-trip lifecycle • Seeded drift surfaces in next memo • Broken-signature promote denied + recorded
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"The wires are set up"

MERGED :

- uds-cli #5026 — in-bundle hash-chained attestation manifest
- pepr #5027 — Λ -floor admission module (0.90 / 0.95 / 0.95)
- #5028 — three Zarf packages + UDS bundle, attestations sidecar wired

LIVE :

- OPA gateway pack — 3 tests, pinned OPA v0.69.0

TRACKED :

- #5118 / #5119 — publish + validate housekeeping (out of scope for Tuesday)

What credibly-auditable agents unlock for the DoD side

- Defensible deployment of LLM-driven decision support inside SCIFs.
- Cross-coalition evidence sharing without releasing the model.
- Insurance-grade posture claims for AI-driven operations.
- A foundation for autonomy the program manager can sign off on.

Risks we've named

License surface — AGPL ↔ Apache managed by dual-licensed upstream PRs.

Doctrine drift — Λ floors payload-anchored; any change is a replay event.

Scope creep — Option C deliberately gated on Andrew's response to A.

Four small asks

- 1. A Mission App target for the Week-3 demo (bland-but-real is best).
- 2. Keycloak realm access on a reference cluster.
- 3. A 30-minute review window at the end of Week 3.
- 4. A thumbs-up to schedule the Option C scoping conversation.

Governed agency

the next primitive the UDS surface deserves —
and we can prove it in three weeks without you taking our word for any of it.